

CSE427 Section 5 Quiz 3

Fall 2025

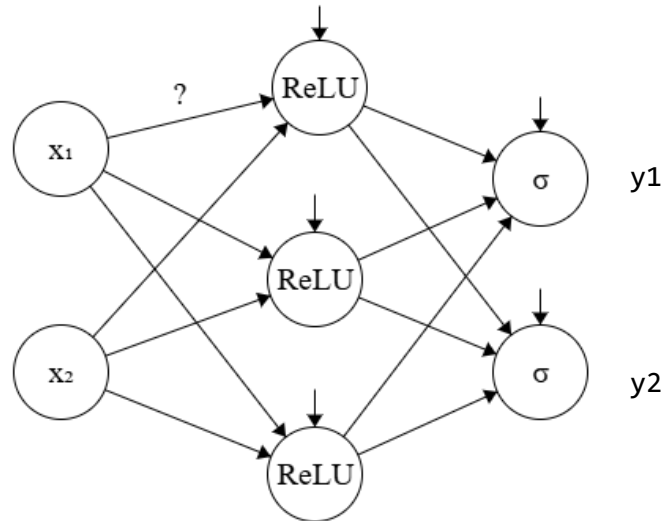
Duration: 25 Minutes

Name :

ID :

Question [10 Points]

Consider the following network:



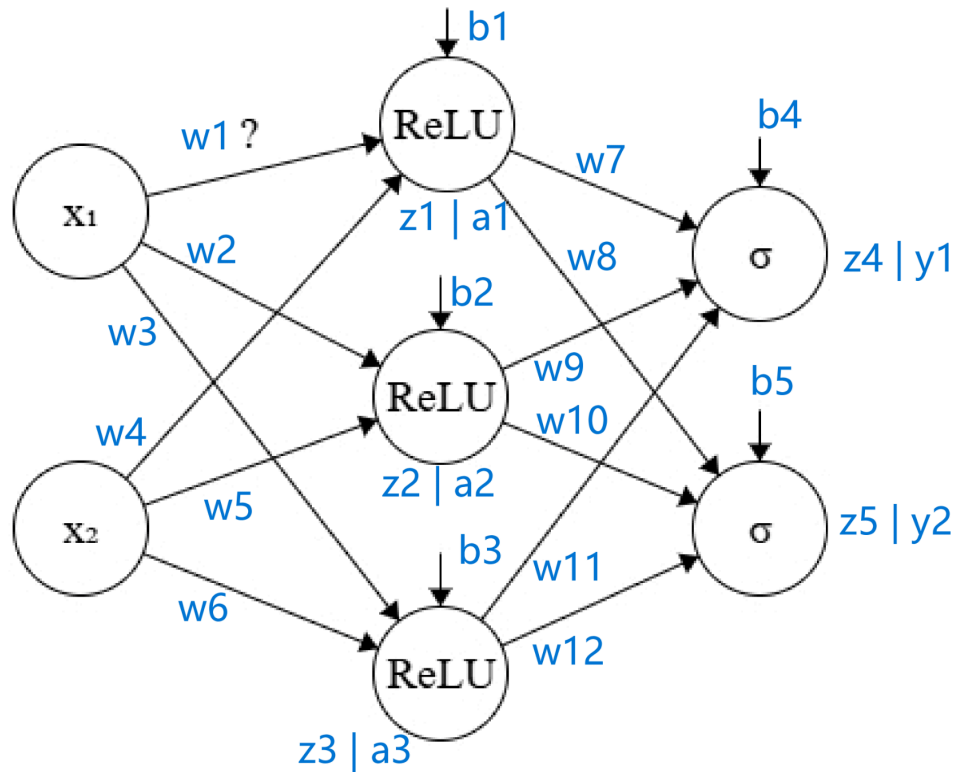
Suppose, all the weights and biases are initialized as 0.5 and 0.1 respectively. Let's assume, you are given the following data:

x1	x2	y1	y2
2	-5	0.4	0.9
-3	8	0.7	0.2

Using the cost function, $E = \text{SSR}/2$; find out the updated “?” marked weight after one forward and backward pass. [Consider, learning_rate = 0.9]

Solution

For the convenience of calculation, I have named the weights, biases, weighted sums and activations like the following:



For the forward pass,

$$z_1 = z_2 = z_3 = z_{123} = 0.5x_1 + 0.5x_2 + 0.1$$

$$a_1 = a_2 = a_3 = a_{123} = \text{ReLU}(z_{123}) = \max(0, z_{123})$$

$$z_4 = z_5 = z_{45} = 0.5a_1 + 0.5a_2 + 0.5a_3 + 0.1$$

$$\hat{y}_1 = \hat{y}_2 = \hat{y} = \sigma(z_{45})$$

x_1	x_2	y_1	y_2	z_{123}	a_{123}	z_{45}	$\hat{y}_1$	$\hat{y}_2$
2	-5	0.4	0.9	-1.4	0	0.1	0.525	0.525
-3	8	0.7	0.2	2.6	2.6	4.0	0.98	0.98

$$\text{Now, } E = \frac{SSR}{2} = \frac{1}{2} \left[SSR_{y_1} + SSR_{y_2} \right]$$

$$SSR_{y_1} = \sum_{i=1}^{m=2} \left(\hat{y}_1^{(i)} - y_1^{(i)} \right)^2; SSR_{y_2} = \sum_{i=1}^{m=2} \left(\hat{y}_2^{(i)} - y_2^{(i)} \right)^2$$

Solution

$$\begin{aligned}\frac{\partial E}{\partial w_1} &= \left(\frac{\partial E}{\partial \hat{y}_1} \times \frac{\partial \hat{y}_1}{\partial z_4} \times \frac{\partial z_4}{\partial a_1} \times \frac{\partial a_1}{\partial z_1} \times \frac{\partial z_1}{\partial w_1} \right) + \left(\frac{\partial E}{\partial \hat{y}_2} \times \frac{\partial \hat{y}_2}{\partial z_5} \times \frac{\partial z_5}{\partial a_1} \times \frac{\partial a_1}{\partial z_1} \times \frac{\partial z_1}{\partial w_1} \right) \\ &= \left(\sum_{i=1}^{m=2} \left(\hat{y}_1^{(i)} - y_1^{(i)} \right) \times \left[\sigma(z_4) \times (1 - \sigma(z_4)) \right] \times w_7 \times ReLU'(z_1) \times x_1^{(i)} \right) \\ &\quad + \left(\sum_{i=1}^{m=2} \left(\hat{y}_2^{(i)} - y_2^{(i)} \right) \times \left[\sigma(z_5) \times (1 - \sigma(z_5)) \right] \times w_8 \times ReLU'(z_1) \times x_1^{(i)} \right)\end{aligned}$$

Note that, if you used Stochastic Gradient Descent instead of Batch Gradient Descent, then the $\sum$ symbol in the beginning won't be there.

If we now plug the values in the above equation, we get:

$$\begin{aligned}\frac{\partial E}{\partial w_1} &= (0.525 - 0.4) \times [0.525 \times (1-0.525)] \times 0.5 \times 0 \times 2 \\ &\quad + (0.98 - 0.7) \times [0.98 \times (1-0.98)] \times 0.5 \times 1 \times (-3) \\ &\quad + (0.525 - 0.9) \times [0.525 \times (1-0.525)] \times 0.5 \times 0 \times 2 \\ &\quad + (0.98 - 0.2) \times [0.98 \times (1-0.98)] \times 0.5 \times 1 \times (-3) \\ &= 0 + (-0.008) + 0 + (-0.023) \\ &= -0.031\end{aligned}$$

$$w_{1(new)} = w_{1(old)} - \eta \times \frac{\partial E}{\partial w_1} = 0.5 - 0.9 \times (-0.031) = 0.5279$$